

IV. QUESTIONNAIRE

A. GENERAL INFORMATION ABOUT THE RESPONDANT:

- Country*: Ireland
- Organization(s) or entity(s) responsible for the preparation of the report*:
 - Name: Department of Further and Higher Education, Research, Innovation and Science and the National Open Research Forum
 - Website: <https://www.gov.ie/dfheris/>
 - Email address*:
 - Phone number*:
 - Please describe the role/mandate of your organization: The Department of Further and Higher Education, Research, Innovation and Science is responsible for policy, funding and governance of the Higher and Further Education and research sectors and for the oversight of the work of the State agencies and public institutions operating in those areas.
- Officially designated contact person(s) who completed this survey:
 - Full name*: Emma Hanrahan
- Position*: Higher Executive Officer, Research Policy and Programmes
 - Email address*:
- Full Name(s) of designated official(s) certifying the report: David Keating, Principal Officer, Research Policy and Programmes
- Brief description of the consultation process established for the preparation of the report:
- Other organization(s) or entit(y)(ies) (including non-governmental) consulted for completing this survey:
 - Name of the organization*: National Open Research Forum
 - Website*: <https://dri.ie/norf/>
 - Email address*
 - Sector: × Public ○ Private ○ Civil Society ○ Other

B. QUESTIONNAIRE FOR MEMBER STATES' REPORTING ON THE IMPLEMENTATION OF THE 2021 RECOMMENDATION

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE

- 1.1 Has the 2021 [Recommendation](#) been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

YES

If yes, please indicate which ministries and or national authorities/entities have been involved in the promotion of the 2021 Recommendation.

The 2021 Recommendation has been promoted as part of the development of the [National Open Research Landscape Report](#) (2021) and the [National Action Plan for Open Research 2022 – 2030](#) (2022).

The process of developing these reports was led by Ireland's National Open Research Forum, a broad group that combines the expertise of representatives from policy, research funding and research performing organisations, the library sector, research infrastructures, enterprise and other key stakeholders in the research system across Ireland. The National Open Research Forum is a Government of Ireland initiative, funded by the Department of Further and Higher Education, Research, Innovation and Science, through the Higher Education Authority and supported by the Royal Irish Academy and the Digital Repository of Ireland.

The *National Action Plan for Open Research* has been officially endorsed by 57 national and international entities, including national ministries, non-departmental state bodies, funders, institutions and affiliated organisations. The relevant ministries are the Department of Further and Higher Education, the Department of Agriculture, Food and Marine, the Department of the Environment, Climate and Communications, and the Open Data Initiative of the Department of Public Expenditure and Reform. Details of endorsing organisations are available here:

<https://norf.ie/national-action-plan/>

Further details of the *National Action Plan for Open Research* are available in answer 2.2.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 1.2 Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

YES.

If yes, which language(s)?

English.

- 1.3 Have there been awareness raising activities on open science, including all its key elementsⁱ, associated benefits and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities? (ref.: [\(i\) 16.f, 16.g, 16.h](#))

YES.

If yes, please provide details on the activities organized or foreseen and links as relevant.

National Open Research Forum Funding for the National Open Research Forum, the entity responsible for driving and implementing Ireland's *National Action Plan for Open Research*, is provided by Department of Further and Higher Education, Research, Innovation and Science, through the Higher Education Authority. Further details are available in answer 2.2.

Elements of Open Science: All elements of open science.

National Action Plan for Open Research 2022-2030 This action plan outlines policies and recommendations to align Irish publicly funded research with open science practices, emphasising establishing a culture of open research, achieving 100% open access to research publications and ensuring FAIR research data and other outputs.

The process of developing this report was informed by close consultation and co-creation with the national research community. Ireland's National Open Research Forum led this process. Further details are available in answer 2.2.

Elements of Open Science: All elements of open science.

National Open Research Fund The Open Research Fund has been a key driver of the *National Action Plan for Open Research*, accelerating Ireland's transition to an open research environment in strategic and targeted ways. Funding is provided by the Department of Further Higher Education Research Innovation and Science, through the Higher Education Authority to distribute to key initiatives identified by the National Open Research Forum. Further details, including details of the funded projects, are available in answer 2.6.

Elements of Open Science: All elements of open science.

A Review of Open Research in Connected Health Ecosystems in Ireland

Regarding 16.g on engaging the private sector, the NORF-funded "A Review of Open Research in Connected Health Ecosystems in Ireland (ROCHE)" project is promoting the adoption of open research principles and policies in public-private research partnerships. Further details are available in answer 2.8.

Elements of Open Science: Open dialogue with other knowledge systems.

National Persistent Identifier Strategy and Roadmap The project to deliver this roadmap was initiated by the National Open Research Forum in response to Ireland's *National Action Plan for Open Research* which includes actions to invest in persistent identifier infrastructure, to support the Irish Open Researcher and Contributor Identifier (ORCID) consortium, and to develop a roadmap, in line with international best practice, to boost the adoption of PIDs. Further details are available here: <https://norf.ie/pid-roadmap/>

Elements of Open Science: Open scientific knowledge, open science infrastructure

National Open Research Festival 2023 The National Open Research Festival, organised by the National Open Research Forum on 1-3 November 2023, at the Royal Irish Academy, showcased a dynamic program focused on all elements of open science. The festival covered topics such as research assessment, data stewardship, open access, FAIR research, and more. Full details of the event are available here: <https://norf.ie/irelands-national-open-research-festival-2023/>

Elements of Open Science: All elements of open science.

EOSC NTE and EOSC-Ireland On 28 September 2022 the first European Open Science Cloud National Tripartite Event took place in Dublin. The event was hosted by the Department of Further and Higher Education, Research, Innovation and Science. The aim of the event was to support increased understanding of and collaboration between respective national and international EOSC activities. Following this NTE EOSC-Ireland, an informal national structure, was established to coordinate between EOSC and national stakeholders.

Elements of Open Science: Open Science Infrastructure

National Research Integrity Forum The National Forum on Research Integrity (NRIF) was established in June 2015. It is coordinated by the Irish Universities Association (IUA) and the Technological Higher Education Association (THEA).

Elements of Open Science: Research Integrity

- 1.4 Have specific actions been undertaken or are planned to be undertaken by end of 2025 in your country to incorporate the values and principles of open scienceⁱⁱ, in publicly funded research? (*ref.:* [\(i\) 16.a, 16.b, 16.c, 16.d, 16.e, 16.h](#))

YES

If yes, please indicate the actions that are taken to incorporate the specific values and principles. For each case, please provide details and links as relevant.

Key actions include:

National Action Plan for Open Research 2022-2030: This action plan outlines policies and recommendations to align Irish publicly funded research with open science practices, emphasising establishing a culture of open research, achieving 100% open access to research publications and ensuring FAIR research data and other outputs.

The process of developing the *National Action Plan for Open Research* was informed by close consultation and co-creation with the national research community. Ireland's National Open Research Forum led this process. Further details are available in answer 2.2.

Open Research Fund 2022 and 2023: The Open Research Fund has been a key driver of the *National Action Plan for Open Research*, accelerating Ireland's transition to an open research environment in strategic and targeted ways. Funding is provided by Department of Further Higher Education Research Innovation and Science, through the Higher Education Authority to distribute to key initiatives identified by the National Open Research Forum. Further details are available in answer 2.6.

Secondary Rights, Copyright, Open access, Institutional Policies and Rights retention (SCOIR): The publication of draft model legislation in the Copyright and Related Rights Bill 2024 by the NORF-funded SCOIR project marks a pivotal step toward ensuring open access to major research outputs produced by researchers and academics at Irish institutions. Further details are available here: <https://scoir.moodlecloud.com/mod/url/view.php?id=114>

Funding Agency Requirements: A summary of current progress made by a selection of the Irish research funding agencies regarding their adoption of open research metrics and supporting policies and processes, is provided in Table 1.

Table 1: Summary of OA/OR practice Adoption by Selected Irish Funding Agencies

Funder	OA/OR Policy and/or Processes?	Dedicated funding available to support OR practices?	Does OR practice feature in Funding Application Assessment?	Does OR practice feature in Project Reporting Requirements?
DAFM	Yes	No	Yes	Yes

EPA	Yes	Yes	No	No (under development)
HEA	Yes	Yes	Yes	Yes
HRB	Yes [DORA Signatory]	Yes	Yes	Yes
Marine Institute	Yes	Yes	Yes	Yes
Research Ireland	Yes [DORA Signatory]	Yes	Yes	Yes
Teagasc	Yes	Yes	No	No

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so?~~

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policyⁱⁱⁱ, strategy or plan of action on science, technology and innovation (STI)?

YES.

If yes

- please attach/upload a machine-readable PDF file of the policy and provide the following information: *Title of the policy, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, and links to the webpage if available.*
- does it include any of the following open science elements:
 - open access to scientific knowledge
 - open science infrastructure
 - open engagement of societal actors
 - open dialogue with other knowledge systems

Title: [Impact 2030: Ireland's Research and Innovation Strategy](#)

Authority in charge of development of the policy: Department of Further and Higher Education, Research, Innovation and Science

Entities involved and the process of development: Details of Governance structures are available here: <https://www.gov.ie/en/publication/27c78-impact-2030-irelands-new-research-and-innovation-strategy/>

Year of adoption: 2022

Year of last update: 2022

Key areas/sections of the policy: The five pillars of Impact 2030 are:

1. Maximising the Impact of Research and Innovation on Our Economy, Society and the Environment.

2. Impact of Research and Innovation Structures on Excellence and Outcomes.
3. Innovation Driving Enterprise Success.
4. Talent at the Heart of the Research and Innovation Ecosystem.
5. All-Island, EU and Global Connectivity.

- does it include any of the following open science elements:

Open Access to Scientific Knowledge	Yes this is being progressed under objectives - 2.5, 4.3 and 5.2.
Open Science Infrastructure	Not specifically but some work on this is being done across the national system.
Open Engagement of societal actors	Yes under objective 1.1.2 Ireland is seeking to strengthen the connections between Government Departments and the public research system and also the citizen's voice in the R&I process.
Open Dialogue with other knowledge systems	Yes under objective 4.2 Inclusion.

Reason:

Impact 2030: Ireland's Research and Innovation Strategy outlines a strong vision for Ireland to become an international leader in open research and positions open research as a key policy agenda.

Within the five pillars of *Impact 2030*, open research falls under **Pillar Two** on the impact of research and innovation structures on excellence and outcomes. As part of good research practice, open research and a positive research culture will support the quality and impact of Irish research.

It also falls under **Pillar Five** on all-Island, EU and Global connectivity within broader objectives relating to the ten-year European Pact for Research & Innovation, the European Research Area Policy Agenda and national ambitions in Horizon Europe.

In this document, the National Open Science strategy is devolved to NORF, and a commitment is made via increased support for Open Science to enable the delivery of the *National Action Plan for Open Research*.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 2.2 In your country, is there a policy, or a set of policies and/or legal framework(s)^{iv} at the national level that address open science or any of its key elements in line with the definition, values and principles outlined in the 2021 Recommendation on Open Science? (*ref.: (i) [16.a](#), [16.b](#), [16.c](#), [16.d](#), [16.e](#), [16.h](#)*)

YES.

If yes or partly, please attach/upload machine-readable PDF file(s) of the policy(ies) and provide the following information for each: *Title of the policy or legal*

instrument, the authority in charge of development of the policy, entities involved and the process of development, year of adoption, year of last update, key areas/sections of the policy, elements of open science that are addressed, and links to the webpage(s) if available.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policies.~~

The [National Action Plan for Open Research 2022 – 2030](#) builds on a number of national policies and international recommendations including the UNESCO Recommendation on Open Science. In the implementation of this plan, particularly in terms of infrastructure, actions should be guided by the principles outlined in the UNESCO Recommendation on Open Science regarding open science infrastructures.

Title: [National Action Plan for Open Research 2022-2030](#)

Authority in charge of development of the policy: National Open Research Forum

Entities involved and the process of development: The process of developing this National Action Plan has been led by Ireland's National Open Research Forum (NORF), a broad group that combines the expertise of representatives from policy, research funding, research performing, the library sector, research infrastructures, enterprise and other key stakeholders in the research system across Ireland. Full details of Forum membership are available here: <https://norf.ie/governance/>

During a planning exercise conducted in 2021, NORF Working Groups and Coordination Groups analysed the landscape of open research support structures in Ireland and developed recommendations for nationally coordinated actions that would improve support for open research and assist researchers in navigating this transition. NORF's analyses have been published in a National Open Research Landscape Report and series of policy briefs that provide background and further details on the actions described in this plan. Opportunities for public consultation and comment were held in conjunction with the drafting of the National Open Research Landscape Report, followed by targeted consultations with specific communities to help develop recommended actions, and a further public consultation on a draft version of the National Action Plan in 2022.

Whilst the National Action Plan for Open Research was being developed, NORF was chaired by the Health Research Board and the Higher Education Authority. Secretariat support was provided by the Department of Further and Higher Education, Research, Innovation and Science.

Funding for the National Open Research Forum is provided by Department of Further and Higher Education, Research, Innovation and Science through the Higher Education Authority.

Year of adoption: 2022

Year of last update: 2022

Key areas of the policy: The three strategic objectives of the National Action Plan for Open Research are:

1. to establish a culture of open research,
2. to achieve 100% open access to research publications,
3. and to enable FAIR research data and other outputs.

Elements of Open Science that are addressed:

Scientific publications	Yes
Open research data	Yes
Open educational resources	Partly
Open source software and source code	No
Open hardware	No
Virtual Open Science Infrastructures	Yes
Physical Open Science Infrastructures	Yes
Citizen and participatory science	Yes
Scientific volunteering	No
Crowdsourcing	No
Crowdfunding	No
Local communities	Yes
Marginalised scholars	Yes
Indigenous peoples	No

2.3 In your country, are there specific policy instruments^v that aim to promote open science?

YES.

If yes or under development, please provide details and links as relevant, including for each instrument: *Title of the instrument, its objective, the authority in charge, allocated budget, elements of open science included.*

Key policy instruments include:

1. National Action Plan for Open Research (2022-2030)

Objective: *To promote open research practices, achieve 100% open access to research publications, and establish the necessary infrastructure and training for open science.*

Authority in Charge: *National Open Research Forum (NORF), under the Department of Further and Higher Education, Research, Innovation and Science (DFHERIS).*

Budget: *€4.33 million allocated by DFHERIS since 2022.*

Elements of Open Science Included: *Open access, FAIR data principles, training programs, national repository systems, and compliance with international frameworks like DORA and the Leiden Manifesto.*

More Information: <https://norf.ie/>

2. Funding Agency Open Access and Open Science Policies

Objective: *To ensure research outputs funded by public agencies are openly accessible, fostering transparency and reusability of research data. Further details are available in answer 1.4.*

3. National Open Access Monitor

Objective: *To track and report Ireland's progress in achieving 100% open access to research outputs. Further details are available in answer 2.9.*

Authority in Charge: *IREL National Open Access Monitoring Advisory Board*

Budget: Pilot funding €199,799.92 provided by the National Open Research Fund.

Elements of Open Science Included: Open access monitoring, transparency in research output dissemination.

More Information: <https://norf.ie/national-open-access-monitoring/>

4. Higher Education Authority (HEA) Principles of Good Practice in Research within Higher Education Institutions

- **Objective:** These nine elements of good practice guide Higher Education Institutions to develop good research practices within their own research environments to help drive research excellence and further improve research outcomes with the view to enhancing the international reputation of the Irish Higher Education Research System.
- **Authority in Charge:** Higher Education Authority (HEA).
- **Budget:** Funded through HEA's broader allocation for higher education research.
- **Elements of Open Science Included:** Integration of open science practices in institutional frameworks, research training.
- **More Information:** <https://hea.ie/>

5. National Policy Statement on Ensuring Research Integrity in Ireland

- **Objective:** To commit the main organisations in Irish research to the highest standards of integrity in carrying out and disseminating their research so that partners, the public and other stakeholders, and the international research community can have full confidence in the Irish research ecosystem.
- **Authority in Charge:** National Research Integrity Forum (NRIF)
- **Budget:** Funding is predominantly provided by the HEA, who provided €100,000 to the NRIF for their 2024 annual budget. In 2020, NRIF received 50% (€150,000) of funding from the HEA to develop and implement Research Integrity training 2021-2023, the other 50% of funding was provided by research performing organisations (RPOs).
- **Elements of Open Science Included:** Open access, open communication, research data management practices and management.
- **More Information:** <https://www.iua.ie/wp-content/uploads/2024/12/National-Policy-Statement-on-Ensuring-Research-Integrity-in-Ireland-Dec-2024.pdf>

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.~~

- 2.4 In your country, is there a national implementation plan, strategy, policy or a roadmap for implementation of open science at the national level in line with the 2021 Recommendation? (ref.: [\(ii\) 17.a, 17.b, 17.c, 17.f, 17.h](#))

YES.

If yes or under development, please provide details and links as relevant, including the name of ministries and national authorities/entities, as well as affiliated organizations that are involved in the development and implementation of the plan, strategy or the roadmap.

Further details are available in answer 2.2.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such policy instruments.~~

- 2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at **institutional** level, including in the context of research-performing institutions, universities, scientific unions and associations and learned societies? (ref.: [\(ii\) 17.d, 17.e, 17.g](#))

YES.

If yes, please provide more information, including the number of policies, name of the institute(s) that has/have developed the policy(s) and [elements of open science](#) included in the policy (please consider policies and strategies addressing all elements of open science, including citizen and participatory science, or co-production of knowledge with multiple actors).

Full details are available in Appendix Two.

- 2.6 In your country, are there specific funding mechanisms^{vi} for open science at the national level? (ref.: [\(ii\) 17.i, \(iii\) 18.i, \(iv\) 19.c](#))

YES.

If yes or under development, please provide details and links as relevant.

Funding Details:

The Open Research Fund has been a key driver of the *National Action Plan for Open Research*, accelerating Ireland's transition to an open research environment in strategic and targeted ways. Funding is provided by the Department of Further and Higher Education, Research, Innovation and Science, through the Higher Education Authority to distribute to key initiatives identified by the National Open Research Forum. The funds are administered by the Digital Repository of Ireland — the National Open Research Forum's host organisation — which is headquartered at the Royal Irish Academy. The Open Research Fund is governed by the National Open Research Forum Grants Oversight Committee. By providing essential funding for the National Open Research Coordinator and Senior Programme Manager roles, the fund has ensured ongoing support and coordination of the National Open Research Forum and its associated initiatives.

Through its strategic investment, the fund has successfully facilitated ten priority action projects that directly address specific goals outlined in the *National Action Plan for Open Research*. These capacity building projects are instrumental in advancing the national agenda. Furthermore, the funding of nine open research stimulus projects has catalysed innovative approaches to open research practices whilst promoting a culture of open research within the research community.

Details of the NORF-funded projects are available in Appendix One and via the NORF website: <https://norf.ie/open-research-fund-projects/>

Additionally, the commissioning of expert analysis into the national Persistent Identifier (PID) landscape has provided critical insights into the current state of open research in Ireland, informing our National Persistent Identifier Recommendations and Roadmap. Details of the PID project, including the final publication are available here: <https://norf.ie/pid-roadmap/>

21/22 €1,725,000

22/23 €1,900,000
23/24/25 €708,000

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to develop such funding mechanisms.~~

- 2.7 In your country, have open science practices^{vii} been integrated in existing research funding mechanisms?

YES.

If yes, please provide details and links as relevant.

See answer 1.4 for further details.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 2.8 Have efforts been made or are foreseen to be undertaken by end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation? (ref.: [\(ii\) 17.i](#))

PARTLY.

A Review of Open Research in Connected Health Ecosystems in Ireland (ROCHE)

Details: The aim of this NORF-funded project is to promote the adoption of open research principles and policies in public-private research partnerships, identify opportunities and needs, and help support the development of an action plan for the sector. The project is due to finish in Q1 2026.

Project Website: [Open Research in Health – A Review of Open Research in Connected Health Ecosystems in Ireland \(ROCHE\)](#)

~~If yes, please provide details and links as relevant.~~

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators on open science included in the national monitoring and evaluation framework for STI policy (ref.: [\(ii\) 17.i, 23.c](#))

UNDER DEVELOPMENT.

If yes or under development, please provide details and links as relevant.

In line with A6.4 of the *National Action Plan for Open Research*, to “establish a national monitoring service for open access and open research more broadly”, in March 2024, the National Open Research Forum, together with the [Irish Research eLibrary \(IREL\)](#), and [OpenAIRE](#) announced the launch of the [National Open Access Monitor](#). Currently in its pilot phase, the National Open Access Monitor is dedicated to improving data quality and expanding functionalities. This innovative platform offers a range of features allowing for the customisation of monitoring, benchmarking, and evaluation, facilitating evidence-based decisions to advance Open Science initiatives.

In line with A6.2.3 of the *National Action Plan for Open Research*, the period 2025-2027 will entail the expansion of the national open access monitor to

encompass further aspects of open research activity and measures relevant to the national context.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

- 3.1 What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.a](#))

According to the latest official data for 2021, Government Expenditure on Research & Development (GERD), as a % of GDP, was 1.11%

Source: [The Research and Development Budget 2021-2022](#)

- 3.2 What is the latest official data for the percentage of your country's population that have access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers. (ref.: [\(iii\) 18.b](#))

According to the latest available data for 2023, 94% of households in Ireland have internet access, with 86% relying on fixed broadband connections.

Source: [Key Findings Internet Coverage and Usage in Ireland 2023 - Central Statistics Office](#)

- 3.3 Are there national research and education networks (NRENs)^{viii} active in the country? (ref.: [\(iii\) 18.c](#))

YES.

If yes, please provide more information and links as relevant, and indicate to which open science actors in your country those networks are accessible.

HEAnet provides high-speed internet and related ICT services to the education and research communities in Ireland. It supports universities, institutes of technology, and other educational institutions, enabling collaboration and access to global research and education networks.

More details: <https://www.heanet.ie/>

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to establish them.~~

- 3.4 Are there national or institutional [open science infrastructures](#)^{ix} in your country? (ref.: [\(iii\) 18.d, 18.e, 18.k, 18.l, 18.m](#))

YES.

If yes or under development, please indicate which type(s) of infrastructure:

- federated information technology infrastructure for open science, including high-performance computing, cloud computing and data storage.
- community managed infrastructures, protocols and standards, for example those that support bibliodiversity and engagement with society
- platforms for exchanges and co-creation of knowledge between scientists and society

- community-based monitoring and information systems to complement national, regional and global data and information systems.

For each infrastructure, please provide links as relevant and more information with regard to their compliance with the 2021 Recommendation on Open Science in terms of inclusivity, accessibility, interoperability, community governance, FAIR (Findable, Accessible, Interoperable, and Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility and Ethics) principles. Please indicate if the infrastructure is non-commercial.

As a key step in the development of the National Action Plan for Open Research, the NORF members prepared the [National Open Research Landscape Report](#) (2021) to provide a snapshot of the current state of support structures for open research in Ireland. The report is structured according to five strategic areas: open access to research publications; enabling FAIR research data; infrastructures for access to and preservation of research; skills and competencies; and incentives and rewards.

Regarding infrastructures for access to and preservation of research, the National Open Research Landscape Report (pp. 55-71) identifies and discusses infrastructure under four headings:

1. Networking and Computing infrastructures
2. Meta-infrastructures
3. Data infrastructures
4. Thematic infrastructures

The inventory of Irish Open Access Repositories is available here: <https://www.universityofgalway.ie/openrepositories/inventory/>

In addition to the research infrastructures discussed in the National Open Research Landscape Report, NORF has funded 10 priority action projects that directly address specific goals outlined in the *National Action Plan for Open Research* and nine open research stimulus projects since 2022, detailed in Appendix One.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 3.5 Does your country host or fund any federated regional or international information technology infrastructure for open science? (ref.: [\(iii\)](#) [18.e](#), [18.f](#))

YES.

If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open science¹, stage of research, accessibility to all the relevant users in the region or internationally).

As a key step in the development of the National Action Plan for Open Research, the NORF members prepared the [National Open Research Landscape Report](#) (2021) to provide a snapshot of the current state of support structures for open research in Ireland. Further details of the National Open Research Landscape Report are available in answer 3.4.

A detailed breakdown of national nodes for some key European Research Infrastructure Consortia (ERICs) and European Strategy Forum on Research Infrastructures (ESFRIs) is provided in Appendix 3 of the [Report](#) (pp. 126-130)

~~If no, are the existing open science infrastructures at regional and international levels accessible to all the relevant open science actors from your country?~~

- 3.6 Is your country involved in any North-South, North-South-South and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional and national open science platforms? ([ref.: \(iii\) 18.g](#))

NO.

~~If yes, please provide links as relevant and more information, including the name of the infrastructure, date of establishment, language, target groups and beneficiaries, governance/community agreements, scope (e.g. research areas, elements of open scienceⁱ, stage of research, accessibility to all the relevant users in the region or internationally).~~

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY AND CAPACITY BUILDING FOR OPEN SCIENCE

- 4.1 Has systematic capacity building on open science taken place in higher education and/or research performing institutions in your country? ([ref.: \(iv\) 19.a, 19.c](#))

YES.

If yes, please indicate if those capacity building initiatives are formal or informal and provide more information and links as relevant.

Each of the ten NORF-funded priority action projects comprise of a formal consortium of national higher education and research performing institutions together with other national and international entities. Details of key NORF-funded activities are listed in Appendix One.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 4.2 Have capacity building programmes for policy makers on how to integrate [core values](#) and [principles](#) of open science in STI policies and strategies taken place in your country?

YES.

If yes, please provide more information and links as relevant.

National Open Research Forum

For further details see answer 2.2.

National Research Integrity Forum

For further details see answer 1.3.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025? ([ref.: \(iv\) 19.b](#))

YES.

If yes, please provide more information and links as relevant.

Aligned with A3.1.1 of the *National Action Plan for Open Research*, the NORF-funded Open Research Training Programme (TROPIC) aims to equip researchers at all career stages, including postgraduate students, with essential open research skills. The TROPIC project unites researchers and librarians from across Ireland's higher education institutions, spanning disciplines in science, engineering, social sciences, and the humanities.

In 2023, the project surveyed researchers, librarians, and research support staff to assess awareness and engagement with open research practices. Insights from the survey, expert input, and disciplinary knowledge inform the development of a curriculum tailored to Ireland's research context. The programme promotes core research values and transferable skills, such as conducting transparent, ethical, and trustworthy research and managing outputs effectively, all aligned with open research best practices.

Following the pilot phase, a second phase will establish a network of open research trainers and foster a community to sustain, enhance, and expand the training materials, ensuring the programme's long-term impact. Further details: <https://osf.io/chxgd/>

~~If no~~, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

- 4.4 Are open educational resources^x as defined in the [2019 Recommendation on Open Educational Resources](#), used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above? (ref.: (iv) 19.d)

YES.

In line with A3.1.4 of the National Action Plan for Open Research, the NORF-funded TROPIC project (detailed in answer 4.3) will establish a national network of open research trainers to promote the exchange of best practices and to develop and share Open Education Resources (OERs) in open research. Further details: <https://osf.io/chxgd/>

- 4.5 Have there been any initiatives in your country to support science communication in line with open science value and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers and the public at large? (ref.: (iv) 19.e)

YES.

If yes, please provide more information and links as relevant.

Campus Engage

Campus Engage is a national initiative, working with a network of 20 higher education institutions, the Irish Universities Association and the Institutes of Technology Ireland, to enable and embed civic and community engagement activity, including community-based research, community-based learning and volunteering, across campuses and communities in Ireland. Further details are available here: <https://www.teachingandlearning.ie/2014/09/15/campus-engage/>

Public Engagement and Open Research

This NORF-funded project will co-create a road map for public engagement with research in Ireland. This will be achieved through a thorough understanding of how public engagement is presented in the literature, as well as through a series of collective intelligence workshops with multiple stakeholders across the academic, public and NGO sectors. The project seeks to build knowledge and capacity but also a network to implement the road map and, thereby, advance the level and quality of public engagement in open research across Ireland. Further details, including details of consortium members, are available here:

<https://norf.ie/engaged/>

HEA Principles of Good Practice in Research within Irish HEIs

The Higher Education Authority's (HEA) Principles of Good Practice in Research within Irish Higher Education Institutions were first launched by the HEA in February 2020 and subsequently revised in December 2022 to take into account recent changes to the Irish research landscape. Open research and the promotion of knowledge transfer are emphasized and encouraged under these principles. The complete set of principles focus on nine elements of practice, which help guide Higher Education Institutions (HEIs) to develop good research practice within their respective research environments. These are as follows:

1. Excellence in research and academic freedom;
2. Research integrity and ethics;
3. Open Research;
4. Intellectual property and knowledge transfer;
5. Researcher development;
6. Research project and programme management;
7. Dignity and respect;
8. Gender equality and inclusiveness, and;
9. Research and sustainability.

Further details are available here: <https://hea.ie/policy/research-policy>

Irish Research e-Library (IReL)

IReL is an essential research infrastructure and enabler of educational activity. It supports access to online information resources to 220,000 students and researchers in universities across Ireland. By providing access to world class online information resources to universities across Ireland IReL creates a level playing field for researchers and students in terms of access to information to support research, innovation and learning. IReL resources had a record 19.1 million accesses in 2022. International benchmarking has proven that IReL delivers value for money to the research sector in Ireland. IReL also increases the impact of Irish research by supporting researchers to publish Open Access (OA), making Irish research more widely available globally and beyond academia. In the three years to date that IReL has received funding for OA (2021-2023), it has supported the publication of almost 10,000 research publications. This ensures publications from IReL member institutions will be immediately available to citizens, students, researchers, and industry.

Further information is available here: <https://irel.ie/about-irel/>

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

- 5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research performing institutions, to review the existing research assessment and evaluation processes in order to align them with open science principles and values or are they foreseen by the end of 2025? (ref.: [\(v\) 20.b, 20c, 20i\)](#)

YES.

If yes, please provide more information and links as relevant.

The National Action Plan for Open Research sets forth the overall national vision for establishing a culture of open research as follows:

“By 2030 a culture of open research will be embedded at every level of the Irish research system. The reorientation of research recognition and rewards towards responsible research evaluation and open research metrics and practices will be well underway or complete, and approaches to research assessment will be broadly aligned at a system level across all stakeholders ... Embedding open research through responsible research evaluation and assessment, national training and education, and increased specialist support, will promote and support the quality and impact of Irish research”.

Addressing A3.2.1 on the incentivisation of open research, A3.3.1 on research assessment and A3.3.2 on responsible research metrics of the *National Action Plan for Open Research*, the NORF-Funded Incentivising Open Research Practices in Ireland, locally known as ABROAD, brings together a strong consortium of research performing organisations, research funding organisations and key researcher groups to engage in deep, iterative dialogue with key stakeholders across the Irish research landscape. The project will use a consultation-driven approach to reach agreement on core commitments and concrete steps to effectively incentivise open research practices, develop a statement on the use of Responsible Research Metrics, and support the development of Responsible Research Assessment. Further details are available here: <https://norf.ie/aboard-project/>

Upon recommendation from NORF's international peer review panel, the Irish National Chapter of CoARA was established under the Incentivising Open Research Practices in Ireland project. This initiative is playing a pivotal role in aligning Ireland with international efforts to improve research culture and integrate responsible metrics into research evaluations. There are currently thirteen national members of CoARA. Technological University Dublin became the first Irish CoARA member to launch its CoARA strategy in September 2024. Further details are available here: <https://researchprofiles.tudublin.ie/en/publications/technological-university-dublin-coara-action-plan-september-2024>

The NORF-funded Responsible Use of Research Metrics project piloted training on responsible use of research metrics (RURM) through an online, accessible module for those involved in recruitment, research assessment, and promotion decisions, co-designed with the research community. The intention is to scale-up as an Open Education Resource (OER). The module will be launched in early 2025.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 5.2 Have specific clauses that recognize open science practices, including sharing, collaborating and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.c, 20.d](#))

YES.

~~If yes, please provide details and links as relevant.~~

For details of the NORF-funded Incentivising Open Research Practices in Ireland project and Ireland's participation in CoARA, see answer 5.1.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025? (ref.: [\(v\) 20.a, 20.b, 20.c, 20.e, 20.f, 20.g, 20h](#))

YES.

~~If yes, for each initiative, please provide more information, including on the stakeholders^{xi} that are involved or lead the initiative, and links as relevant.~~

For details of the NORF-funded Incentivising Open Research Practices in Ireland project and Ireland's participation in CoARA, see answer 5.1.

In addition, Lero's Annual Director's Awards includes an award category of Open Science Champion to reward the researcher who has contributed the most in the previous year to developing Lero's Open Science culture. Lero's [Open Science Charter](#) is formulated in-line with the eight pillars of the EU's Open Science Policy. Further details are available in Appendix Two.

In 2023, NORF's [Open Research Fund](#) sought to incentivise researchers from all disciplines who wish to play a leading role in advancing open research in Ireland via the Open Research Stimulus Call 2023. For further details see answer 2.6.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 5.4 Have there been any unintended negative consequences^{xii} of open science practices reported in your country? (ref.: [\(v\) 20](#))

NO.

~~If yes, please provide more information and specify if there is a strategy or plan to preventing and mitigating such consequences.~~

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

- 6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages

of the research cycle to increase the openness of the entire scientific process? (ref.: [\(vi\) 21](#))

YES, at both national and institutional levels.

If yes, please indicate which of the recommended actions they address and provide more information and links as relevant:

- promotion of pre-prints
- exploration of open peer-review practices
- valuing and sharing of negative scientific results and associated data
- participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects
- development of participatory strategies for identifying the needs of marginalized communities and highlighting socially relevant issues to be incorporated into research agendas.
- promoting open innovation practices.
- Others (with a box to add text)

Key initiatives include:

National Action Plan for Open Research 2022-2030 The overall vision of the National Action Plan for Open Research is for a research system fully aligned with open research principles and practices by 2030. Further details are available in answer 2.2.

Recommended actions: promoting innovative approaches for open science at all stages of the scientific process.

Health Research Board (HRB) Open Research Platform is a platform for HRB-funded researchers to rapidly publish their research outputs in an open and accessible way. Further details are available here: <https://hrbopenresearch.org/about>

Recommended actions: promotion of pre-prints, exploration of open peer-review practices

Star Guide: Showing Citizens the Best View of the Exoplanets The NORF-funded Star Guide project worked with international partners and affiliates to make exoplanet observation more accessible. International networks of citizen scientists work around the clock to refine and update exoplanet transit timings. Star Guide supports their efforts by making accessible, user-friendly observation planning systems. Further details are available here: https://github.com/creanero/star_guide

Recommended actions: citizen science, crowdsourcing scientific projects

Public Engagement and Open Research Further details are available in answer 4.1.

Recommended actions: development of participatory strategies.

Managing Open Source Software in Irish Universities (MOSS-I) The NORF-funded MOSS-I project has developed an Open Source Software policy framework for Irish Universities. Further details are available here: <https://norf.ie/moss-i/>

Recommended actions: promoting open innovation practices, user-friendly access to open source software

A Review of Open Research in Connected Health Ecosystems in Ireland (ROCHE) Further details are available in answer 2.8.

Recommended actions: open innovation

Open Access Repository Assessment and Alignment project and Open Research Ireland Further details are available in answer 4.1.

Recommended actions: Developing strategies that facilitate the deposit of data in archives

ERIC Membership A detailed breakdown of national nodes for some key European Research Infrastructure Consortia (ERICs) and European Strategy Forum on Research Infrastructures (ESFRIs) is provided in Appendix 3 of the [National Open Research Landscape Report](#) (pp. 126-130)

Recommended actions: open access of large-scale research infrastructures

European Open Science Cloud (EOSC) Ireland has also established synergies with broader infrastructures such as the European Open Science Cloud (EOSC) via representation in EOSC Association and expert contributions to EOSC Working Groups

Recommended actions: open access of large-scale research infrastructures.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

7. **PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS**

- 7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science? (ref.: [\(vii\) 22.a\)](#))

In line with A6.1.1 of the National Action Plan for Open Research, the National Open Research Coordinator represents NORF and presents on Open Research developments in Ireland nationally and internationally including through participation in the Council for Open Science Coordination (CoNOSC) and the Informal Commission Expert Group on Scientific Information.

Irish institutions actively participate in the Research Data Alliance (RDA), which works to establish global, community-managed standards and infrastructures for research data sharing. The Digital Repository of Ireland (DRI) is the RDA-Ireland Node.

- 7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regards to exchange of best practices, capacity building and metrics for open science? (ref.: [\(vii\) 22.b](#), [22.e](#), [22.f](#))

YES.

If yes, please provide links as relevant and more information.

[Interoperability, openness, and impact: Recommendations and roadmap for an Irish national PID strategy](#), presents 15 key recommendations and detailed actions aimed at establishing a more efficient, transparent, and open research ecosystem in Ireland, spanning the period until 2030. The objective is to enable

seamless information flow between national and international systems and organisations, reduce bureaucratic burdens, and enhance research transparency and integrity. PIDs underpin this vision as they act as critical bridges between entities.

~~If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.~~

- 7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science? (ref.: [\(vii\) 22.c](#))

NO.

~~If yes, please provide links as relevant and more information.~~

If no, please, if possible, briefly explain the reason and the difficulties or obstacles in this area and indicate if there are plans to do so.

8. GENERAL CONSIDERATIONS

- 8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

YES.

If yes, please provide more information.

The potential impact of growing international concerns about data security on Open Science are detailed in the following open letter by the international Open Science Collaborative: <https://opensciencecollaborative.substack.com/p/coming-soon>.

Appendix One: NORF-Funded projects

NOTE REGARDING COMPLIANCE WITH THE 2021 RECOMMENDATION ON OPEN SCIENCE:

For all NORF-funded projects, it is a condition of funding that Open Research practices should be followed and identified; a Data Management Plan is required as part of the full application, along with a plan for updating it. It is required that all solutions are open (i.e., no vendor lock-in), that the dissemination of project outputs are Open Access, and that project data are managed and shared in compliance with the FAIR principles. Each of the NORF-funded projects has a strong community-led governance structure.

NATIONAL OPEN ACCESS MONITOR

Funding Call: 2022 Open Research Fund – Priority Action

Open Science Infrastructure Type: Digital infrastructure for open science; Community-Based Monitoring and Information Systems

Description: Led by IReL, the National Open Access Monitor project is a two-year project funded under the National Open Research Forum 2022 Open Research Fund. This project will develop a fully-open monitor for Open Access (OA) at the national level to publish, analyse and track progress towards the national target of 100% OA to research publications by 2030.

Compliance: Ensures inclusivity and accessibility. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs, FAIR alignment, community-driven governance, open source.

Non-Commercial: Yes

Link: <https://irel.ie/oamonitor/>

OPEN ACCESS REPOSITORY ASSESSMENT AND ALIGNMENT PROJECT AND OPEN RESEARCH IRELAND

Funding Call: 2022 Open Research Fund – Priority Action

Open Science Infrastructure Type: Services supporting the open science approach; Community-Managed Infrastructures, Protocols, and Standards

Description: Led by the University of Galway the Open Access Repository Assessment and Alignment Project aims to deliver standardised metadata nationally while building and aligning infrastructure according to international best practice. Based on the ongoing work of this project, Open Research Ireland is a new membership organisation dedicated to advancing open research practices and repository services across Ireland. Further details, including details of consortium members, are available here: <https://norf.ie/oa-repository-assessment-and-alignment/>

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability, and FAIR/CARE alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/oa-repository-assessment-and-alignment/>

OPEN ACCESS TRANSITION PROGRAMME

Funding Call: 2022 Open Research Fund – Priority Action

Open Science Infrastructure Type: Digital infrastructure for open science; Community-Managed Infrastructures, Protocols, and Standards

Description: Led by the Royal Irish Academy and Trinity College Dublin, this project aims to conduct a feasibility study and pilot with a view to establishing a publicly owned, centralised national platform for Diamond OA publication of journal and books. This action is aimed at supporting OA models for Irish-based academic journals and publishers. Further details, including details of consortium members, are available here: <https://norf.ie/oa-transition-programme/>

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability, and FAIR alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-commercial: Yes

Link: <https://publishoa-ie.moodlecloud.com/>

OPEN RESEARCH TRAINING PROGRAMME (TROPIC)

Funding Call: 2022 Open Research Fund – Priority Action

Open Science Infrastructure Type: Services (network) supporting the open science approach

Description: The aim of the project led by Maynooth university is to develop and pilot a national open research training programme to upskill researchers in the fundamentals of open research practices. Following the pilot phase, a second phase will establish a network of open research trainers and foster a community to sustain, enhance, and expand the training materials, ensuring the programme's long-term impact. Further details are available in answer 4.3.

Compliance: Ensures inclusivity, accessibility, and FAIR/CARE alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-commercial: Yes

Link: <https://osf.io/chxgd/>

NATIONAL DATA STEWARDSHIP STEWARDSHIP NETWORK (SONRAÍ)

Funding Call: 2022 Open Research Fund – Priority Action

Open Science Infrastructure Type: Services supporting the open science approach; Community-Managed Infrastructures, Protocols, and Standards

Description: Led by the University College Cork, Sonraí is a network of and for data stewards to support Ireland's transition to a FAIR and open research data landscape. Further details, including details of consortium members, are available here: <https://norf.ie/national-data-stewardship-network/>

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability and FAIR/CARE alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: <https://datastewards.ie/>

SHARED DATA STORAGE SERVICE PILOT (IRLDAT)

Funding Call: 2022 Open Research Fund – Priority Action

Open Science Infrastructure Type: Federated information technology infrastructure for open science

Description: The pilot will enable users to access a national shared storage infrastructure and a suite of data management services during the active phase of the research data life cycle to verify need for such services, their fitness for purpose and will conclude with a report for policymakers and funders defining a sustainable national scale production service.

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability and FAIR alignment. Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/shared-data-storage-service-pilot/>

Incentivising Open Research Practices in Ireland (ABOARD)

Funding Call: 2023 Open Research Fund – Priority Action

Open Science Infrastructure Type: NA

Description: Led by University College Cork and Trinity College Dublin, this project will use a consultation-driven approach to reach agreement on core commitments and concrete steps to effectively incentivise open research practices, develop a statement on the use of Responsible Research Metrics, and support the development of Responsible Research Assessment. Further details of the Incentivising Open Research Practices in Ireland project are available in answer 5.1.

Compliance: NA

Non-Commercial: NA

Link: <https://norf.ie/aboard-project/>

Public Engagement and Open Research (ENGAGED)

Funding Call: 2023 Open Research Fund – Priority Action

Open Science Infrastructure Type: Platform for exchanges and co-creation of knowledge between scientists and society; services (networks) supporting the open science approach.

Description: This project will co-create a road map for public engagement with research in Ireland. This will be achieved through a thorough understanding of how public engagement is presented in the literature, as well as through a series of collective intelligence workshops with multiple stakeholders across the academic, public and NGO sectors. The project seeks to build knowledge and capacity but also a network to implement the road map and, thereby, advance the level and quality of public engagement in open research across Ireland.

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability and FAIR/CARE alignment; Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/engaged/>

Irish National Research Data Management Framework (iFRAME)

Funding Call: 2023 Open Research Fund – Priority Action

Open Science Infrastructure Type: Services supporting the open science approach, Community-Managed Infrastructures, Protocols, and Standards

Description: The iFrame project's aim is to develop a national research data framework that will help institutions to upgrade their RDM service provision and to align with best practices. The framework will also be a stepping stone for the organisation of national support services regarding technical service provision and in access to discipline specific expertise.

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability and FAIR/CARE alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/iframe/>

Open Access Policies and Rights Retention

Funding Call: 2023 Open Research Fund – Priority Action

Open Science Infrastructure Type: NA

Description: The project will adopt a two-pronged approach to policy and legislative change. To develop a secondary publishing right in Irish legislation and adoption of institutional rights retention policies. Such institution's policy will allow the author's accepted version of a research output be shared via an institutional repository. The Legislation will be drafted in support of secondary publication rights based on an analysis of international and Irish law. It will also develop an open access policy framework for both funders and institutions that is aligned with the National Action Plan for Open Research and international best practice.

Compliance: NA

Non-Commercial: NA

Link: <https://scoir.moodlecloud.com/>

STAR GUIDE: SHOWING CITIZENS THE BEST VIEW OF THE EXOPLANETS

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: Platforms for exchanges and co-creation of knowledge between scientists and society

Description: The NORF-funded Star Guide project worked with international partners and affiliates to make exoplanet observation more accessible. International networks of citizen scientists work around the clock to refine and update exoplanet transit timings. Star Guide supports their efforts by making accessible, user-friendly observation planning systems.

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability and FAIR/CARE alignment. Community-driven governance.

Non-Commercial: Yes

Link: https://github.com/creanero/star_guide

QSA*NET: PROMOTING OPEN RESEARCH IN QUALITATIVE SOCIAL SCIENCES

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: Services (networks) supporting the open science approach, Community-Managed Infrastructures, Protocols, and Standards

Description: QSA*Net aims to stimulate awareness and participation in open research by promoting practices of data re-use in the qualitative social sciences. The project will research barriers to re-using qualitative data deposited in data infrastructures and inform researchers about the analytical potential of Qualitative Secondary Analysis (QSA). It will engage researchers across disciplines, and across academic and applied sectors, by building community, fostering expertise, and developing skills in QSA, with a particular focus on early career and other under-represented researchers. QSA*Net will establish an international partnership on excellence in QSA in Ireland and in the UK. Building on this partnership, QSA*Net will develop an openly available collection of state-of-the art resources on and will contribute to the development of an international network on Qualitative Archives and Qualitative Data Re-Use.

Compliance: Ensures inclusivity, accessibility, international interconnectivity and interoperability and FAIR/CARE alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/gsanet/>

CULTIVATING AN OPEN RESEARCH CULTURE IN THE HUMANITIES AND SOCIAL SCIENCES

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: Services (networks) supporting the open science approach

Description: The overall objective of the project is to foster a culture of open research in the humanities and social sciences (HSS) by building an open research network of researchers and practitioners (HSS); providing training in open research practices; understanding the tensions between open research and research assessments; and, exploring the rewards and recognition systems in supporting open research, including funding programmes and awards

Compliance: Ensures inclusivity, accessibility, and FAIR alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/cultivating-an-or-culture/>

EMBEDDING A CULTURE OF INTERDISCIPLINARY OPEN RESEARCH IN IRELAND'S CRIMINAL JUSTICE SECTOR

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: Services (networks) supporting the open science approach

Description: The project involves the establishment of the Criminal Justice Open Research Dialogue (CORD) Partnership. The CORD Partnership includes

representatives from partner organisations across research, criminal justice policymaking, agencies and oversight bodies, non-state justice services, civil society and advocacy groups, and the broader research ecosystem. In 2024, partners met three times at workshops that use restorative practices and design thinking approaches to establish the principles, purposes, priorities and actions to embed a culture of interdisciplinary open research in criminal justice in Ireland.

Compliance: Ensures inclusivity, accessibility, and FAIR/CARE alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: <https://norf.ie/open-research-in-criminal-justice/>

PILOT TRAINING FOR RESPONSIBLE USE OF RESEARCH METRICS

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: NA

Description: The project proposes pilot training on the Responsible Use of Research Metrics (RURM) through an online, accessible module for those involved in recruitment, research assessment, and promotion decisions, co-designed with the research community.

Compliance: NA

Non-Commercial: NA

Link: <https://norf.ie/responsibleuserresearchmetrics/>

A REVIEW OF OPEN RESEARCH IN CONNECTED HEALTH ECOSYSTEMS IN IRELAND

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: NA

Description: The goal of A Review of Open Research in Health Ecosystems in Ireland (ROCHE) is to understand policies, practices, challenges, and opportunities for open research in commercialisable research, particularly in Digital/Connected Health research and innovation in Ireland. Through interviews with stakeholders and policy analysis, ROCHE will benchmark the state of the art in Ireland against comparator countries; investigate knowledge and practice in both publicly-funded research performing organisations and the private sector; and provide guidance for funders, research and innovation support staff, and other stakeholders.

Compliance: NA

Non-Commercial: NA

Link: <https://norf.ie/connected-health-ecosystems/>

BUILDING A CULTURE OF OPEN RESEARCH FOR IRISH HEALTH AND SOCIAL CARE PRACTITIONER RESEARCHERS

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: Digital infrastructure for open science, Community-Managed Infrastructures, Protocols, and Standards, Services (networks) supporting the open science approach

Description: This project aims to build on these established foundations of Open Research support in a way that is understanding of the challenges of health practitioner research. It will formally establish an Open Health Research network to bring together this researcher cohort. It also aims to provide viable routes to realising the goal of 100% open access through the development of an Open Access Journal for Irish health and social care research.

Compliance: Ensures inclusivity, accessibility and FAIR alignment. Prioritises bibliodiversity by supporting diverse forms of scholarly outputs. Community-driven governance.

Non-Commercial: Yes

Link: Not Available

MANAGING OPEN SOURCE SOFTWARE IN IRISH UNIVERSITIES (MOSS-I)

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: NA

Description: The MOSS-I project has developed a clear framework dedicated to guiding users in the research and academic community on the uses of Open Source Software (OSS). The framework (available here: <https://zenodo.org/records/14392733>) focuses on:

- Creating awareness for users about the background and uses of Open Source Software.
- Explaining the various types of OSS licences.
- Highlighting the potential impacts and considerations for using OSS effectively with guidance from the Technology Transfer Office (TTO) in Ireland.

It is designed to promote uniformity in advice provided by Irish universities; this framework will serve as a reliable resource for understanding, consuming, contributing to, and creating OSS. It will support the Irish research system to strengthen, promote or better support open source practices within open research practices. It will incorporate guides and references from globally recognised university sources, who offer in-depth knowledge about OSS principles, practices, processes, and tools, all contextualised within an Irish framework.

Compliance: NA

Non-Commercial: NA

Link: Not Available

Sharing Arts Practice FAIRly: Developing Best Practice Training and Guidance to Support the FAIR Dissemination of Arts Practice Research Outputs

Funding Call: 2023 Open Research Fund – Open Research Stimulus Call

Open Science Infrastructure Type: NA

Description: This project aims to support and develop best practice guidance and training for the FAIR dissemination of arts practice research outputs. Its methods include a case study on the use of a new open access repository by arts practice researchers at University of Limerick, and the compilation of a field report on current dissemination practices in three additional Irish HEIs (Trinity College Dublin, University College Cork and University of Galway). Using this work as a foundation, it will produce technical training support and best practice guidelines for the FAIR dissemination for arts practice research outputs.

Compliance: NA

Non-Commercial: NA

Link: <https://norf.ie/sharing-arts-practice-fairly/>

National Persistent Identifier Strategy and Roadmap

Funding Call: 2022 Open Research Fund – Consultant Led Project

Open Science Infrastructure Type: NA

Description: The project to deliver this roadmap was initiated by the National Open Research Forum in response to Ireland's *National Action Plan for Open Research* which includes actions to invest in persistent identifier infrastructure, to support the Irish Open Researcher and Contributor Identifier (ORCID) consortium, and to develop a roadmap, in line with international best practice, to boost the adoption of PIDs. Further details are available here: <https://norf.ie/pid-roadmap/>

Compliance: NA

Non-Commercial: NA

Link: <https://norf.ie/pid-roadmap/>

Appendix Two: Institutional Policies and/or Strategies that Promote Open Science in line with the 2021 Recommendation

The *National Action Plan for Open Research 2022-2030* (p.5) states that: “At an organisational level, we encourage the prioritisation of open research in institutional strategies, supported by policies and organisational action plans for open research aligned with international and national objectives. Policies and procedures should cover key areas including open access, research data management, and research assessment. Institutional support networks should be broad, involving professional support staff as well as teams of academic champions or fellows that support local awareness, engagement and effective implementation.”

Institution	Endorses NAP	OR in institutional strategies	Open Research Policy/ Strategy	Open Access Policy/ Strategy*	RDM Policy/ Strategy**	Other Initiatives including Research Assessment
Publicly funded universities						
University College Dublin	Yes	No	No	No – Compliance with funders’ requirements.	Yes	CoARA Member Agape: An introductory course to OS for ECRs UCD Research Culture Initiative Pilot Training for Responsible Use of Research Metrics – Project Lead Cultivating an Open Research Culture in the Humanities and Social Sciences – Project Lead
University College Cork	Yes	Yes	No	Yes – included in UCC Strategic Plan	Yes	CoARA Member DORA signatory ABOARD: System-level Incentivisation of Open Research Practices – Project Lead Sonraí – Irish Data Stewardship Network – Project Lead
University of Galway	Yes	No	No	Yes	Yes	CoARA Member DORA signatory Open Access Repositories

						Assessment and Alignment – Project Lead Open Repositories Ireland Open Research Forum (Local) Open Scholarship Community Galway
Maynooth University	Yes	Yes	No	Yes	No MU provides guidance on research data management through its RDM LibGuide	DORA signatory Open Research Training Programme – Project Lead Maynooth Open Research Network Criminal Justice Open Research Dialogue – Project Lead
University of Limerick	Yes	Yes – Research Strategy 2022-2027	No	No – Compliance with funders' requirements.	Yes	CoARA Member DORA signatory Irish National Research Data Management Framework – Project Lead Sharing Arts Practice FAIRly – Project Lead
Trinity College Dublin	Yes	Yes	No – Open communication and open scholarship included in the Policy on Good Research Practice 2024	Yes	Yes – Policy on Good Research Practice 2024	CoARA Member DORA signatory Open Scholarship Group Open Access Transition Programme – Co-lead Incentivising Open Research Practices in Ireland (ABOARD) – Co-lead Public Engagement and Open

						Research (ENGAGED) – Co-lead Open Access Policies and Rights Retention (SCOIR) – Co-lead Building a Culture of Open Research for Irish Health and Social Care Practitioner Researchers – Co-lead
Dublin City University	Yes	Yes	No – implementing the NAP via the Open Research Steering Group	No – Compliance with funders' requirements.	No DCU provides guidance on research data management through its RDM LibGuide	CoARA Member DORA signatory Open Research Steering Group Star Guide: Showing Citizens the Best View of the Exoplanets – Project Lead
Technological University Dublin	Yes	Yes	Yes via TU-Net Joint Statement on Open Research	Yes	Yes via TU-Net Joint Statement on Open Research	CoARA Member - CoARA Action Plan DORA signatory Full details of TUD's Open Research initiatives are available here: https://www.tudublin.ie/explore/news/tu-dublins-open-research-successes-lead-the-way.html
Munster Technological University	Yes	No	Yes via TU-Net Joint Statement on Open Research	Yes TU-Net Joint Statement on Open Research KPIs included in Strategic Plan	Yes TU-Net Joint Statement on Open Research	

Technological University of the Shannon	Yes	No	Yes via TU-Net Joint Statement on Open Research	Yes via TU-Net Joint Statement on Open Research	Yes via TU-Net Joint Statement on Open Research	CoARA Member DORA signatory
Atlantic Technological University	Yes	No	Yes via TU-Net Joint Statement on Open Research	Yes via TU-Net Joint Statement on Open Research	Yes via TU-Net Joint Statement on Open Research	Open Scholarship Community Galway
South East Technological University	Yes	Yes	Yes via TU-Net Joint Statement on Open Research	Yes via TU-Net Joint Statement on Open Research	Yes via TU-Net Joint Statement on Open Research	CoARA Member DORA signatory
Institutes of Technology						
Dundalk Institute of Technology	Yes	No	No	Yes	Yes	
Dun Laoghaire Institute of Art, Design and Technology	No	No	No	No IADT provides guidance on open access through its Open Access LibGuide .	No publicly available information confirming an institutional research data management policy.	
Other Institutions that Receive Public Funding						
Marino Institute of Education	No	No	No	No OA support included in Research Strategy 2023-2028	No MIE provides guidance on RDM through its RDM LibGuide	
Mary Immaculate College	Yes	No	No	No OA support included in MIC	No MIC provides guidance on	CoARA Member

				Strategic Plan 2023 .	research data management through its Research Data Management LibGuide.	
National College of Art and Design	No	No	No	No publicly available information confirming an institutional open access policy.	No publicly available information confirming an institutional research data management policy.	
National College of Ireland	In Progress	No	No	No NCI provides guidance on open access through its Open Access LibGuide	No publicly available information confirming an institutional research data management policy	
Pontifical University of Maynooth	No	No	No	No publicly available information confirming an institutional open access policy	No publicly available information confirming an institutional research data management policy	
St Patrick's, Carlow College	No	No	No	No publicly available information confirming an institutional open access policy.	No publicly available information confirming an institutional research data management policy.	

Royal Irish Academy of Music	No	No	No	No publicly available information confirming an institutional open access policy.	No publicly available information confirming an institutional research data management policy.	
Royal College of Surgeons Ireland	Yes	Yes	No	Yes	Yes	DORA signatory
Royal Irish Academy	Yes	Yes	No	Yes. The RIA has agreements facilitating open access publishing for staff and students of participating institutions. Information can be found on the RIA website .	No publicly available information confirming an institutional research data management policy.	National Open Access Transition Programme – Co-lead The National Open Research Forum, the entity responsible for the NAP, is headquartered under the RIA
Dublin Institute for Advanced Studies	Yes	No	No	Yes	No publicly available information confirming an institutional research data management policy.	
Other Examples						
Lero – Research Center for Software	Lero's Open Science Charter is formulated in-line with the EU's Open Science policy , which identifies the following eight Open Science ambitions: • Open Data • European Open Science Cloud (EOSC) • New Generation Metrics • Future of Scholarly Communication					

	• Rewards & Incentives • Research Integrity • Education and Skills • Citizen Science
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* The NORF-funded Institutional Open Access Policies and Rights Retention Project members are working to create/review, strengthen and support institutional Open Access policies in line with the objectives of the [National Framework on the Transition to an Open Research](#), the National Action Plan for Open Research and international exemplars.

** The NORF-funded National Research Data Management Framework is developing a national research data framework that will help institutions to upgrade their RDM service provision and to align with best practices.

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- i Please refer to the elements of open science defined in the 2021 Recommendation, namely : [open scientific knowledge](#) (including open access to scientific publications, research data, open educational resources, open-source software and code, open hardware), [open science infrastructures](#), [open engagement of societal actors](#) and [open dialogue with other knowledge systems](#).
- ii The 2021 Recommendation outlines a set of shared values of open science stemming from the rights-based, ethical, epistemological, economic, legal, political, social, multi-stakeholder and technological implications of opening science to society and broadening the principles of openness to the whole cycle of scientific research. It also provides a set of guiding principles, as a framework for enabling conditions and practices, which uphold the values of open science and can realize the ideals of open science.
Please see the [values of open science](#), namely quality and integrity, collective benefit, equity and fairness, and diversity and inclusiveness.
Please see the [guiding principles for open science](#), namely transparency, scrutiny, critique and reproducibility; equality of opportunities; responsibility, respect and accountability; collaboration, participation and inclusion; flexibility and sustainability.
- iii National STI policy is most commonly a governmental document developed by governing bodies leading the STI policy design and implementation in a given country, which formulates objectives and guides actions and decision-making processes in the area of STI on the national level.
- iv These can include specific national policies, sets of guidelines, rules, regulations, laws, principles, or directions to govern open science practices.
- v Policy instruments include programmes, methods and mechanisms of technical and operational nature, required to solve the issues set by the policy, with focus on the target beneficiaries, resources, indicators, and set of goals for delivering products (short term), results (medium-term), and impacts (long term).
- vi A funding mechanism refers to a policy instrument or process through which financial resources are allocated or provided for a specific purpose. Some examples are block funding, research grants, support for technological poles and centres of excellence, support for science parks and innovation centres, Infrastructure grants (for research facilities, labs, instruments), communication and outreach funds, innovation funds, loans and tax credits scholarships, studentships, fellowships, trust funds, sectoral funds.
- vii Examples can include sharing open scientific knowledge (publications, research data, educational resources, software and hardware) with other scientists and with the public, sharing or using open infrastructures, collaboration with other researchers, with national and local societal actors such as policy makers, or sectors such as industry, agriculture, and health, and collaboration and open dialogue with indigenous peoples and local communities.
- viii A national research and education network (NREN) is a specialised internet service provider dedicated to supporting the needs of the research and education communities within a country.
- ix Some examples of open science infrastructures include major shared scientific equipment or sets of instruments, open computational and data manipulation service infrastructures that enable collaborative and multidisciplinary data analysis, open science platforms and repositories for publications, research data and source codes, archives, open bibliometrics and scientometrics systems for assessing and analysing scientific domains, open laboratories, software forges and virtual research environments, open innovation testbeds, incubators, science museums, science parks and infrastructure for non-digital materials.
- x Open Educational Resources are learning, teaching and research materials in any format and medium that reside in the public domain or are under copyright that have been released under an open license, that permit no-cost access, re-use, re-purpose, adaptation and redistribution by others ([2019 Recommendation on Open Educational Resources](#))

xi For example, research funders, universities, research institutions, learned societies, scientific publishers and journal editorial boards.

xii For example, predatory behaviours, data migration, exploitation and privatization of research data, increased costs for scientists and high article processing charges associated with certain business models in scientific publishing that may be causes of inequality for the scientific communities around the world and, in some cases, the loss of intellectual property and knowledge.